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Computer Science Department

Software Engineering

Advanced Web Development

NRC: 30716

Team: Code & Coffe

Members: Carlos Alexander Torres Pincay, Evelyn Haydée Villarreal Salgado

Instructor: Engineer Edison Lascano

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ALCSys^{tem} v2.0.1

RESTful URI Design Document

1. Document Purpose

This document describes the URI design currently implemented in the American Latin Class web system. It documents the existing backend API routes and frontend page routes.

The document follows an academic URI design structure: system context, base URLs, RESTful design rules, backend endpoint catalog, frontend page catalog, examples, authentication, validation, and error conventions.

2. System Overview

ALCSystem is a web platform for a dance academy under the name American Latin Class. The system includes public pages for external visitors and protected portals for students, teachers, and directors.

- Norte
- Matriz
- Tumbaco
- Quitumbe
- Conocoto

Area	Description
Public website	Presents the academy, pricing, offers, and enrollment request form.
Authentication	Allows student, teacher, and director users to access the internal portal.
Student portal	Shows student profile, progress, attendance, schedule, and events.
Teacher portal	Shows attendance, worked hours, planning documents, and payroll-related records.
Director portal	Manages students, teachers, payroll, attendance, planning, finance, and B2 events.
Teacher attendance station	Registers teacher check-in from the academy computer.

3. Architecture Context

Layer	Context	Responsibility
Model	06Code/Model	Database entities managed with Eloquent ORM.
View	06Code/View	Static frontend pages deployed with Netlify.
Controller	06Code/Controller/src/Controller	Slim 4 controllers that expose backend API routes.
Validation	06Code/Controller/src/Service/Validation	Input validation services used by controllers.
Database	Supabase PostgreSQL	Production database.
Backend deployment	Render	PHP backend API service.
Frontend deployment	Netlify	Static frontend hosting and dashboard rewrites.

4. URI Count Summary

Side	Implemented count	Notes
Backend	27 route entries	1 root route and 26 API routes under /api.
Frontend	16 canonical route entries	5 public or operational pages and 11 dashboard routes.

5. Base URLs

Environment	Base URL
Production backend	https://american-latin-class.onrender.com
Production frontend	https://american-latin-class-frontend.netlify.app
Local backend	http://127.0.0.1:8080
Local frontend	http://127.0.0.1:5500

6. RESTful URI Design Rules Used

Rule	Application in ALCSysSystem
Use nouns for resources	/api/students, /api/teachers, /api/branches
Use plural collection names	students, teachers, class-plans, attendance-records
Use path parameters for specific resources	/api/students/{studentId}
Use query parameters for filters	/api/students?branch_id=1&level=B2
Keep API routes under one prefix	/api
Return JSON from backend endpoints	All controller responses use JSON.
Protect internal resources	Protected routes use Authorization: Bearer <token>.

7. Authentication and Authorization

Protected endpoints use Bearer token authentication.

```
Authorization: Bearer <token>
```

The token is obtained through the login endpoint:

```
POST /api/auth/login
```

Role	Main access
student	Student profile and personal monthly attendance.
teacher	Class plans, attendance records, teacher work logs, and attendance control views.
director	Student management, teacher management, payroll, planning, finance, and B2 events.

8. Backend API Endpoint Catalog

#	Method	URI	Auth	Roles	Purpose
1	GET	/	No	Public	Backend project metadata and endpoint summary.
2	GET	/api/health	No	Public	Checks API and database connection status.
3	GET	/api/branches	No	Public	Lists academy branches.
4	POST	/api/enrollments	No	Public	Creates a public enrollment request.
5	POST	/api/auth/login	No	Public	Authenticates a user and returns a token.
6	POST	/api/kiosk/attendance	No	Public station	Registers legacy student kiosk attendance.
7	POST	/api/teacher-attendance/check-in	No	Public station	Registers teacher check-in from academy computer.
8	GET	/api/me	Yes	student, teacher, director	Returns the authenticated user profile.
9	GET	/api/me/attendance	Yes	student	Returns student monthly attendance.
10	GET	/api/students	Yes	director	Lists students with optional filters.
11	POST	/api/students	Yes	director	Creates a student record.
12	PATCH	/api/students/{studentId}	Yes	director	Updates a student record.
13	DELETE	/api/students/{studentId}	Yes	director	Deactivates a student record.
14	GET	/api/teachers	Yes	director	Lists teacher accounts.
15	POST	/api/teachers	Yes	director	Creates a teacher account.
16	PATCH	/api/teachers/{teacherId}	Yes	director	Updates a teacher account.
17	DELETE	/api/teachers/{teacherId}	Yes	director	Deactivates a teacher account.
18	GET	/api/class-plans	Yes	teacher, director	Lists class planning records.
19	POST	/api/class-plans	Yes	teacher, director	Submits a class planning record.
20	GET	/api/attendance-records	Yes	teacher, director	Lists attendance and payroll summary.
21	POST	/api/attendance-records	Yes	teacher, director	Creates a manual attendance record.
22	GET	/api/branch-finance-reports	Yes	director	Lists branch finance reports.
23	POST	/api/branch-finance-reports	Yes	director	Creates a branch finance report.
24	GET	/api/professional-events	Yes	director	Lists professional B2 events.
25	POST	/api/professional-events	Yes	director	Creates a professional event.
26	POST	/api/professional-events/{eventId}/assignments	Yes	director	Assigns a B2 dancer to an event.
27	GET	/api/dancer-settlements/{studentId}	Yes	director	Calculates B2 dancer settlement summary.

9. Backend Endpoint Details

9.1 Root and Health

GET / returns backend metadata, framework information, and available endpoint groups.

```
{
  "project": "American Latin Class Backend API",
  "framework": "Slim 4",
  "architecture": "MVC controllers with Eloquent models",
  "database": "Supabase PostgreSQL",
  "health": "/api/health"
}
```

GET /api/health checks if the API and database are available.

```
{
  "status": "ok",
  "database": "connected",
  "project": "American Latin Class"
}
```

9.2 Public Enrollment

POST /api/enrollments creates a public enrollment request with pending status.

```
{
  "branch_id": 1,
  "national_id": "1723456789",
  "full_name": "Valeria Paz",
  "email": "student@example.com",
  "phone": "0990000000",
  "level": "B1",
  "scholarship_percent": 0,
  "guardian_name": "Luis Paz",
  "guardian_phone": "0991112222",
  "comments": "Interested in urban dance classes."
}
```

Field	Rule
branch id	Required and must reference an existing branch.
national id	Required, 10 digits, valid Ecuadorian ID.
full name	Required, letters only, maximum 120 characters.
email	Required, valid email, maximum 254 characters.
phone	Required, 7 to 20 characters after sanitization.
level	Must be B1 or B2.
scholarship percent	Must be 0, 25, 50, 75, or 100.

9.3 Authentication

POST /api/auth/login authenticates a student, teacher, or director account.

```
{
  "email": "director@americanlatinclass.com",
  "password": "ALC2026*"
}
```

```
{
  "token": "jwt-token-value",
  "user": {
    "id": 1,
    "email": "director@americanlatinclass.com",
    "role": "director",
    "name": "Juan Pablo Hidalgo"
  }
}
```

9.4 Attendance Stations

POST /api/teacher-attendance/check-in registers a teacher check-in from the academy computer.

```
{
  "email": "teacher@americanlatinclass.com",
  "branch_id": 1,
  "expected_start_time": "18:00",
  "duration_hours": 1,
  "style": "Salsa",
  "notes": "Regular class."
}
```

9.5 Students and Teachers

The student and teacher resources are protected director modules.

Resource	Implemented operations
Students	GET /api/students, POST /api/students, PATCH /api/students/{studentId}, DELETE /api/students/{studentId}
Teachers	GET /api/teachers, POST /api/teachers, PATCH /api/teachers/{teacherId}, DELETE /api/teachers/{teacherId}

9.6 Class Plans

GET /api/class-plans lists planning records. POST /api/class-plans submits a planning record.

```
{
  "branch_id": 1,
```

```

"teacher_name": "Andrea Molina",
"month": "2026-05",
"level": "B2",
"objective": "Improve salsa partner work.",
"activities": "Warm-up, footwork, partner routine.",
"document_url": "https://example.com/class-plan.pdf"
}

```

9.7 Attendance Records and Payroll

GET /api/attendance-records lists attendance records and returns teacher payroll summary.

POST /api/attendance-records creates a manual attendance record.

Field	Accepted values or format
person type	student, teacher
status	present, absent, late, excused
attendance date	YYYY-MM-DD
expected start time	HH:MM
duration hours	Between 0.25 and 8

9.8 Finance and Professional Events

Directors can manage finance reports and B2 professional event records.

Module	Endpoints
Branch finance reports	GET /api/branch-finance-reports, POST /api/branch-finance-reports
Professional events	GET /api/professional-events, POST /api/professional-events
B2 event assignments	POST /api/professional-events/{eventId}/assignments
Dancer settlements	GET /api/dancer-settlements/{studentId}

10. Frontend URI Catalog

Production frontend base URL:

```
https://american-latin-class-frontend.netlify.app
```

#	URI	Access	Purpose
1	/	Public	Home page with academy presentation.
2	/pricing.html	Public	Pricing, dance styles, and offers.
3	/enrollment.html	Public	Enrollment request form.
4	/login.html	Public	Internal portal login.

#	URI	Access	Purpose
5	/attendance-kiosk.html	Academy computer	Teacher check-in station.
6	/dashboard	Protected	Dashboard entry route.
7	/dashboard/overview	Protected	Role-based overview.
8	/dashboard/students	Protected	Student management or student profile module.
9	/dashboard/teachers	Protected	Teacher management or teacher work module.
10	/dashboard/payroll	Protected	Teacher payroll summary.
11	/dashboard/planning	Protected	Class planning module.
12	/dashboard/finance	Protected	Branch finance module.
13	/dashboard/events	Protected	B2 professional events module.
14	/dashboard/schedule	Protected	Student schedule module.
15	/dashboard/attendance	Protected	Student attendance calendar or attendance records.
16	/dashboard/work-log	Protected	Teacher work log module.

10.1 Frontend File Aliases and Rewrites

URI	Type	Notes
/index.html	File alias	Same public home experience as /.
/dashboard.html	File alias	Static dashboard file.
/dashboard/*	Netlify rewrite	Rewrites to /dashboard.html with status 200.
/dashboard/techers	Compatibility alias	Accepted typo route; canonical route is /dashboard/teachers.

11. Frontend Page Details

Page	URI	Connected backend endpoint
Home	/	Public navigation only.
Pricing	/pricing.html	Redirects offers to /enrollment.html.
Enrollment	/enrollment.html	POST /api/enrollments.
Login	/login.html	POST /api/auth/login.
Teacher attendance station	/attendance-kiosk.html	POST /api/teacher-attendance/check-in.
Dashboard	/dashboard/*	Role-based API modules.

12. Error Response Conventions

Status	Meaning	Example
400	Bad request	Invalid JSON or malformed request.
401	Unauthorized	Missing token or invalid credentials.
403	Forbidden	User role cannot access the resource.
404	Not found	Resource was not found.
422	Validation error	Input fields did not pass validation.
500	Server error	Unexpected backend error.


```
{
  "errors": {
    "email": "A valid email is required.",
    "national_id": "National ID is not a valid Ecuadorian ID."
  }
}
```

13. Security Notes

Security area	Current implementation
Password handling	Passwords are hashed in the backend and are never returned as plain text.
Token auth	Protected API routes require a Bearer token.
Role checks	Internal routes use role middleware for student, teacher, and director.
Input validation	Backend validators protect enrollment, student, teacher, attendance, class plan, finance, and event inputs.
Session cleanup	The frontend removes stored session data on sign out.
Frontend cache	Netlify headers disable browser cache for frontend files.

14. Conclusion

ALCSystem currently includes a functional backend API with 27 implemented route entries and a frontend with 16 canonical route entries. The API supports public enrollment, authentication, role-based access, student and teacher management, attendance control, teacher payroll summaries, class planning, branch finance reports, and B2 professional event management.

The URI design documented here represents the implemented business scope of the system at ALCSystem v2.0.1.